

MATTHEW HEMA

Mechanical Engineering Student · 3D Modeling & Hardware Prototyping
Computer Vision and Artificial Intelligence

CONTACT

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Nationality: Thai
Age: 19

EDUCATION

B.Eng, Engineering Product Development
SUTD (ASEAN Scholarship)
Major in Mechanical Engineering
Sep 2025 – Present

High School Diploma
Mahidol Wittayanusorn School
Full Scholarship, GPAX 3.95
May 2022 – Mar 2025

IELTS
7.5 overall, all bands 7.0+

SAT
Math 800

TECHNICAL SKILLS

CAD and Electronics
Prototyping
Project Management
Computer Vision

LANGUAGES

English (Fluent)
Thai (Fluent)
Chinese (Learning)

INTERESTS

Basketball, guitar, videography

AWARDS / ACTIVITIES

- Winner, SUTD Competitive Programming Contest (Jul 2026)
- Exchange Student, Chongqing University, China (Aug 2026)
Exchange Student, Zhejiang University, China (Sep 2026)
- Member, SUTD Basketball Team
- Content Creator, SUTD Basketball Club & Association of Thai Students in Singapore - social media and promotional publications

SUMMARY

Mechanical engineering student passionate about CAD and computer vision, with a strong drive to build AI-integrated hardware. Experienced in medical imaging research, and always keen to learn new tools and technologies.

WORK EXPERIENCE

Research Assistant, Neuroscience Center for Research and Innovation 2024 – 2025
King Mongkut's University of Technology Thonburi

- MRI-based classification of dementia stages using deep convolutional networks
- Real-time sign language translation system using Long Short-Term Memory (LSTM) neural networks

PROJECTS

DayCharge: Solar Cell Bicycle Light 2026

- Rebuilt a salvaged bike-light circuit into a self-charging solar power system, integrating a solar panel and TP4056 charging module for continuous daytime recharging
- Designed and 3D-modeled a custom enclosure in Onshape, precision-fit to the solar panel and engineered for secure handlebar mounting

LazerGraft (Air Draw): Mid-Air Drawing System Using Hand-Landmark Tracking 2026

- Built a real-time computer vision tool enabling mid-air drawing via webcam, replicating the “light board” effect used by online educators
- Implemented hand-landmark tracking with MediaPipe and real-time svtroke rendering with OpenCV

Solar Food Preservation Device 2026

- Co-designed a solar-powered dehydration device for food preservation, handling both the mechanical/CAD design and the solar electronics system
- Built to extend food shelf life without reliance on the electrical grid, using solar heat to drive the drying process

EDUBin: AI-Powered Recycling Educational Tool 2025

- Designed an interactive recycling bin that recognized items and guided students with real-time feedback
- Trained a deep learning classifier and integrated the electronics for automatic sorting

Agrisense: A 2-in-1 Innovative Apparatus for Chemical Detection in Agricultural and Environmental Applications 2024

- Designed a device utilizing the colorimetric method and engineering principles to evaluate the concentration of phosphate and nitrate in soil and water
- Awarded Gold Award, IDEX 2024, Singapore
- Awarded Gold Medal, World Invention Competition and Exhibition (WICE) 2024, Malaysia
- Accepted publication in the Journal of Food Agricultural Sciences and Technology (FAST)

Verdian-01: Vegetation Remote Detection & Integrated Agricultural Satellite 2024

- Designed a CUBE-Satellite concept to monitor crops and environmental conditions remotely